Lecture 5

Time, and the symmetries that forbid superpositions

Objectives. *Time translations supply more structure than separate symmetries. Their phases can be coordinated, by a cited theorem, into the unitary group of QM-II. A lower energy bound then excludes an operator that translates energy by arbitrary amounts. Rotations raise a different question: which rays belong to one coherent sector, and which observables can compare them?*

- State the continuity and lifting premises behind $e^{-itH/\hbar}$, and track its phase freedom.
- Prove Pauli's obstruction with domains, and explain why a bare commutator is insufficient.
- Derive the rotation-sector restriction, represent the states of its observable algebra, and compute a neutron's 4π interference period.

5.1 From ray translations to a Hamiltonian

Lecture 4 classified each symmetry separately. A time evolution must also compose consistently: evolving for s, then t, is evolving for $s+t$. We assume a nonzero separable complex $\mathcal{H}$, with dimension at least two. The one-dimensional case has only one ray and no nontrivial ray dynamics.

Definition 5.1 (Continuous group of ray symmetries)**.** A family $\mathcal{S}_t$, $t \in \mathbb{R}$, consists of ray symmetries and obeys $\mathcal{S}_0 = \mathrm{id}$ and $\mathcal{S}_{s+t} = \mathcal{S}_s\mathcal{S}_t$. Continuity means that $t \mapsto p(\mathcal{S}_t[\psi], [\varphi])$ is continuous for every pair of rays, with the transition probability of Lecture 4.

This is continuity of every pure-state comparison probability, rather than differentiability of a wavefunction. By the half-time square argument, every member has unitary type. Choose representatives $\mathcal{V}_t$. Their products induce the same ray maps as $\mathcal{V}_{s+t}$, so uniqueness in Wigner's theorem gives

$$\mathcal{V}_s\mathcal{V}_t = c(s,t)\mathcal{V}_{s+t}, \qquad |c(s,t)| = 1.$$

This is a *projective* group law: the residual scalar is invisible on rays. Even continuous choices need not have $c = 1$, as Lecture 4's quadratic phase example showed. What is needed is a simultaneous choice of phases compatible with every addition of times.

Theorem 5.2 (One-parameter lift, cited). *Every continuous group of ray symmetries as above has a strongly continuous unitary group U_t implementing it: $\mathcal{S}_t[\psi] = [U_t\psi]$, $U_{s+t} = U_sU_t$, $U_0 = \mathbb{1}$, and $\|U_t\psi - U_{t_0}\psi\| \to 0$ as $t \to t_0$ for every ψ. Any two such group lifts differ by e^{ict} with one real constant c.*

This is Moretti, Theorem 12.45, pp. 714–715; continuity is his Definition 12.40, p. 708. Its proof uses tools beyond our prerequisites and is cited, not derived from Wigner alone. The stated result concerns the real one-parameter group.

QM-II proved that a strongly continuous unitary group has a self-adjoint generator (Theorem 7.8, Stone); its exponential correspondence was sketched there (Theorem 7.10). In that convention the generator is $K = H/\hbar$. Consequently, given the cited lift and that correspondence,

$$U_t = e^{-itH/\hbar}, \qquad H = H^\dagger \text{ on } D(H).$$

This deduction has those stated inputs; it is not a fresh proof of the lifting theorem or the general spectral exponential. Multiplying the lift by e^{ict} replaces H by $H - \hbar c\mathbb{1}$, with the same domain. The constant has units of inverse time, so $\hbar c$ is an energy shift. Ray probabilities cannot fix an absolute energy origin. They also do not select a particular Hamiltonian: the chosen ray dynamics are a premise.

There are three distinct assumptions in this chain. Transition probabilities determine the kind of each symmetry by Wigner. The group law relates distinct times, and probability continuity rules out abrupt ray motion. The lifting theorem then coordinates phases, after which Stone identifies the generator. Dropping the group law permits driven families whose composition depends on the starting time; dropping probability continuity removes the lift's hypothesis, and with it the strong continuity Stone needs. Neither closed-system unitarity nor a time-independent Hamiltonian describes every reduced system: tracing out a partner will lead to different evolution in L7. Thus the theorem explains the form of a specified closed-system dynamics under its premises, rather than assuming every time-indexed state change has this form.

For a density operator write $\rho = \sum_j p_j \left|\varphi_j\right\rangle\!\left\langle\varphi_j\right|$ as in L1. The evolved vectors are still an orthonormal family, so

$$\rho_t = U_t\rho U_t^\dagger = \sum_j p_j \left|U_t\varphi_j\right\rangle\!\left\langle U_t\varphi_j\right|$$

is positive and has trace one. For every bounded A, absolute convergence of these expectation series gives

$$\mathrm{Tr}(\rho_t A) = \sum_j p_j \left\langle U_t\varphi_j, AU_t\varphi_j\right\rangle = \mathrm{Tr}(\rho U_t^\dagger A U_t). \tag{5.1}$$

Thus the Schrödinger and Heisenberg descriptions of QM-I, Proposition 4.5, agree for all bounded readings. No cyclic trace rule for arbitrary unbounded products is needed. Differentiating this identity for an unbounded H requires extra domain conditions; the identity itself holds for every density operator.

Example 5.3 (Two energies and one energy origin). For $H = \mathrm{diag}(\varepsilon_0, \varepsilon_1)$ and $\psi = (|0\rangle + |1\rangle)/\sqrt{2}$, the evolved ray is represented by $(|0\rangle + e^{-it(\varepsilon_1-\varepsilon_0)/\hbar}\,|1\rangle)/\sqrt{2}$. Its probability for the initial ray is $[1 + \cos((\varepsilon_1 - \varepsilon_0)t/\hbar)]/2$. Adding $b\mathbb{1}$ to H changes only the removed common phase $e^{-itb/\hbar}$. The energy difference, and hence the oscillation period, is unchanged by any common energy shift.

Self-check 5.1

1. Does continuity of arbitrary representatives establish the exact group law?
2. If $U'_t = e^{ict}U_t$, what is H'? Which domain changes?

5.2 What Pauli's obstruction actually excludes

A position translation shifts position by any real amount. Could a self-adjoint time operator similarly translate energy? Specify the proposed operation before using the formal expression $[T, H] = i\hbar\mathbb{1}$.

Theorem 5.4 (Pauli, energy-covariant form)**.** *Let H be self-adjoint on a nonzero $\mathcal{H}$ and bounded below: $\langle\psi, H\psi\rangle \geq E_{\min}\|\psi\|^2$ on $D(H)$. There is no self-adjoint T whose unitaries $W_s = e^{isT/\hbar}$ obey, for every real energy s,*

$$W_s D(H) = D(H), \qquad W_s H W_s^{-1} = H - s\mathbb{1}. \tag{5.2}$$

Proof. Choose a unit $\psi \in D(H)$; such a vector exists because a self-adjoint operator is densely defined. Domain invariance puts $\eta = W_s^{-1}\psi$ in $D(H)$, and unitarity gives $\|\eta\| = 1$. Thus

$$E_{\min} \leq \langle\eta, H\eta\rangle = \left\langle\psi, W_s H W_s^{-1}\psi\right\rangle = \langle\psi, H\psi\rangle - s.$$

The fixed expectation on the right is finite. Taking s larger than $\langle\psi, H\psi\rangle - E_{\min}$ contradicts the inequality. ∎

The argument actually excludes any family of energy-shifting unitaries with these properties. It does not need $\psi \in D(T)$ and never differentiates an unbounded product. Self-adjointness of T would supply unitaries for all real s; the lower bound prevents their proposed action. With $U_t = e^{-itH/\hbar}$, covariance would imply the Weyl relation $W_s U_t = e^{ist/\hbar}U_t W_s$. To justify this without differentiating a product on an unspecified common domain, fix s. The conjugated time group has Hamiltonian $W_s H W_s^{-1} = H - s\mathbb{1}$ by its strong derivative on $D(H)$. The group $e^{ist/\hbar}U_t$ has that same Hamiltonian and domain, so QM-II, Proposition 7.9, identifies them. The plus sign also follows from $e^{-it(H-s)/\hbar} = e^{ist/\hbar}e^{-itH/\hbar}$. Formally differentiating covariance at zero yields $[T, H] = i\hbar\mathbb{1}$, but reversing this implication needs hypotheses not contained in a commutator on a common test domain.

Here is a concrete warning. On $L^2(0, L)$ take the self-adjoint twisted momentum P_θ recalled in L4 and a constant $\alpha > 0$ with units energy per length. Set $H = \alpha Q$, $T = -P_\theta/\alpha$. The bounded multiplication operator H is nonnegative; T is self-adjoint conditional on that recalled momentum result (QM-II, Theorem 3.10, sketched there). For $\psi \in C_c^\infty(0, L)$ both products exist and differentiation gives

$$[T, H]\psi = i\hbar\psi.$$

Nevertheless (5.2) cannot hold. Here $D(H) = L^2(0, L)$, so its invariance is automatic. It is the energy-shift identity that fails: unitary conjugation preserves positivity, whereas $H - s\mathbb{1}$ has negative expectations for sufficiently large s. A further boundary warning concerns the different domain $D(T)$: multiplication by x need not preserve it, since $x\psi(x)$ has endpoint values $0, L\psi(L)$. The compact-support commutator controls neither the global energy-shift identity nor every operator domain.

This theorem is about a self-adjoint, globally energy-covariant time. It is not a prohibition of recording arrival times or waiting times. Time POVMs, a generalised reading introduced in

L9, can be covariant; see Busch–Lahti–Pellonpää–Ylinen, p. 390 and Theorem 17.1, p. 395. The familiar short-time survival law of QM-II also refers to elapsed time as a parameter and requires no operator satisfying (5.2).

Self-check 5.2

1. In the proof, where is domain invariance used and where is the lower bound used?
2. Why does the compact-support commutator not contradict the theorem?

5.3 Rotations and coherent sectors

Time translations have a group lift. Rotations illustrate a different possibility. QM-I, Theorem 7.3, gives for each finite spin multiplet a rotation by 2π as $(-1)^{2j}\mathbb{1}$: plus for integer j, minus for half-integer j. Each is a common phase within its own multiplet, hence acts trivially on its rays, as the identity rotation in ordinary three-dimensional space must do.

Call a subspace *coherent for a projective rotation action* if all its rays are admitted as pure states and the ordinary rotation group $SO(3)$ acts on them with a well-defined projective action. Here $SO(3)$ means the ordinary proper rotations of space; "projective" again allows one common phase for each implementing operator, not a separate phase for each component of a vector. We assume that rotations about z are implemented, up to that common phase, by the spin operators $e^{-i\theta J_z/\hbar}$ of those multiplets. This connects the ray action to the stated physical angular momentum; an unrelated abstract rotation action would not suffice. Integer and half-integer blocks may be orthogonal sums of finite multiplets: their unitary rotations act blockwise and extend by norm limits of finite component sums. Thus the 2π signs remain $+1$ and -1 on the respective whole blocks.

Suppose one such subspace contained nonzero integer and half-integer components u, v. They are orthogonal spin sectors. Their sum transforms under 2π as $u + v \mapsto u - v$. These are different rays: if $u - v = a(u + v)$, comparison of the orthogonal components gives both $a = 1$ and $a = -1$. But 2π is the identity element of $SO(3)$ and must fix every ray. This proves the *univalence restriction*: one coherent sector carrying the stated projective action cannot contain both types.

The vector $u+v$ certainly exists in the direct-sum Hilbert space. The conclusion depends on which rays and which group action are physically admitted. Using the group of spin rotation operators instead changes the premise: in that group the 2π spin rotation is not the identity. A group name without its action on states is insufficient to establish a superselection rule. The spin rotation group $SU(2)$ distinguishes a 2π rotation from its identity; a 4π rotation is its identity. Admitting a projective action of the ordinary group $SO(3)$ on every physical ray is therefore a substantive premise. The action within each multiplet alone does not establish an empirical absence of coherence between the two spin types. That absence requires a specification of which preparations and observations are physically available. To describe the restricted observations explicitly, we now specify an algebra rather than silently discarding the Hilbert-space vectors.

Self-check 5.3

1. Why can a common phase not turn $u - v$ into $u + v$ when both components are nonzero?

5.4 A state belongs to an observable algebra

Let $\mathcal{H} = \mathcal{H}_{\text{int}} \oplus \mathcal{H}_{\text{half}}$, with both blocks nonzero and separable, and write their orthogonal projections as $E_{\text{int}}, E_{\text{half}}$. Define the bounded sign operator $\Gamma_s = E_{\text{int}} - E_{\text{half}}$, and specify the observable algebra

$$\mathcal{A} = \{A \in \mathcal{B}(\mathcal{H}) : A\Gamma_s = \Gamma_s A\} = \mathcal{B}(\mathcal{H}_{\text{int}}) \oplus \mathcal{B}(\mathcal{H}_{\text{half}}). \tag{5.3}$$

An *algebra* here is a collection containing the identity and closed under sums, complex scalar multiples, products and adjoints. These properties follow immediately from commutation with the self-adjoint Γ_s. Since $E_{\text{int}} = (\mathbb{1}+\Gamma_s)/2$ and $E_{\text{half}} = (\mathbb{1}-\Gamma_s)/2$, a commuting A preserves both blocks, proving the second equality; every block-diagonal A conversely commutes. Bounded observables are the self-adjoint members of $\mathcal{A}$. For an unbounded observable its spectral projections must belong to $\mathcal{A}$; merely a formal commutator is not its definition.

A state on $\mathcal{A}$ is positive, complex-linear, normalised and normal, as in L1, with orthogonal projection sums now taken in $\mathcal{A}$. There are no allowed cross-block observations in this specification. A pure state means an extreme point of the convex set of states of $\mathcal{A}$, as in L2 with this specified state set. That physical restriction is the *superselection rule*; positivity alone did not force us to choose this algebra.

Theorem 5.5 (Normal states of the two-block algebra). *Every state of $\mathcal{A}$ is represented uniquely among block-diagonal density operators by*

$$\rho = w_{\text{int}}\sigma_{\text{int}} \oplus w_{\text{half}}\sigma_{\text{half}}, \qquad w_j \geq 0, \quad w_{\text{int}} + w_{\text{half}} = 1.$$

Here a σ_j is a density operator on block j when $w_j > 0$; its choice is immaterial when $w_j = 0$. Conversely every such operator defines a normal state on $\mathcal{A}$ by $\omega(A) = \operatorname{Tr}(\rho A)$. The pure normal states are precisely the rays in a single block.

Proof. Put $w_j = \omega(E_j)$. Positivity and $E_{\text{int}} + E_{\text{half}} = \mathbb{1}$ give the claimed weights. If $w_j > 0$, restrict to operators supported on that block and divide by w_j. This is a positive normalised functional on $\mathcal{B}(\mathcal{H}_j)$; orthogonal projection sums there are also such sums in $\mathcal{A}$, so it is normal. L1 supplies a unique σ_j. If $w_j = 0$, then for a positive operator B supported on that block, $0 \leq B \leq \|B\| E_j$ gives $0 \leq \omega(B) \leq 0$. Any bounded self-adjoint operator supported there is a difference of positive ones by adding a multiple of E_j; any bounded operator is a complex combination of two self-adjoint ones. Thus its expectation is zero. Splitting any A into its two blocks proves the representation, and the sum of the two positive traces is one.

Conversely the two density expectation rules are normal by L1, and their weighted sum is normal on $\mathcal{A}$. The weights are recoverable as $\omega(E_j)$, and the positive-weight restrictions determine σ_j; hence the block-diagonal ρ is unique. If both weights are positive, the state is a nontrivial convex mixture of two distinct sector states, so is not pure. If one weight is one, any convex decomposition into states of $\mathcal{A}$ must have zero weight in the other block, by positivity. Purity then reduces to purity on the surviving full block algebra, which L2 identifies with a ray. ∎

For unit $u \in \mathcal{H}_{\text{int}}$, $v \in \mathcal{H}_{\text{half}}$ and $0 < w < 1$, consider $\psi = \sqrt{w}\,u + e^{i\beta}\sqrt{1-w}\,v$. Every $A = A_{\text{int}} \oplus A_{\text{half}}$ has

$$\langle \psi, A\psi \rangle = w \langle u, A_{\text{int}} u \rangle + (1-w) \langle v, A_{\text{half}} v \rangle = \operatorname{Tr}((w\,|u\rangle\langle u| \oplus (1-w)\,|v\rangle\langle v|)A)\,.$$

The relative phase β disappears because all cross-block matrix elements vanish. The ray and the block mixture are the same state *on this algebra*. They are distinguishable on the

full $\mathcal{B}(\mathcal{H})$, by the bounded self-adjoint operator $X_\beta = e^{-i\beta}\,|u\rangle\langle v| + e^{i\beta}\,|v\rangle\langle u|$. Its mean is $2\sqrt{w(1-w)}$ in the ray and zero in the mixture, for every stated phase; X_β is excluded from $\mathcal{A}$. L2's pure-ray theorem concerned the full algebra; restricting observations can turn its pure state into a mixed state. The pair "states and allowed observables" must travel together when specifying a quantum system.

Example 5.6 (A sector weight is not a probability amplitude)**.** Take $\mathcal{H}_{\text{int}} = \mathbb{C}$ and $\mathcal{H}_{\text{half}} = \mathbb{C}^2$. Specify $w_{\text{int}} = 1/3$, $\sigma_{\text{half}} = \mathbb{1}_2/2$. The unique block-diagonal representative is $\rho = \operatorname{diag}(1/3, 1/3, 1/3)$. For an allowed observable $A = a \oplus (b\mathbb{1}_2 + \mathbf{c}\cdot\boldsymbol{\sigma})$, with real coefficients, its expectation is $a/3 + 2b/3$. The Pauli terms average to zero because the state within the half-integer block is mixed. The event E_{half} has probability $2/3$; conditioned on that sector the normalised state is $\mathbb{1}_2/2$, not $(2/3)\mathbb{1}_2/2$. The weights describe classical uncertainty between sectors; the density operators describe the remaining uncertainty within them.

Replace the half-block state by $|\uparrow\rangle\langle\uparrow|$. The resulting state is still mixed on $\mathcal{A}$, even though each nonzero block state is pure: reading the sector projection distinguishes its two components with probabilities $1/3, 2/3$. It is the same restricted state as the Hilbert-space ray $\sqrt{1/3}\,u + e^{i\beta}\sqrt{2/3}\,\uparrow$, for every β. Changing the phase cannot change any allowed expectation. Removing the algebra restriction would restore observations sensitive to that phase, so it would describe a different physical system.

Self-check 5.4

1. On $\mathbb{C}\oplus\mathbb{C}$, find the block density operator for $(|0\rangle + i\,|1\rangle)/\sqrt{2}$. Is its restricted state pure?
2. Which projection probabilities recover the two sector weights?

5.5 A relative 2π rotation is measurable

A neutron has spin one half. If a 2π rotation changes only a common phase, why do neutron interferometers detect the sign? The operation is relative to a second path. Use QM-I's finite tensor product, with path first and spin second. Prepare

$$\Psi_0 = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |\uparrow\rangle\,, \qquad R_z(\theta) = e^{-i\theta\sigma_z/2}.$$

The separate path operations are shown in Figure 5.1.

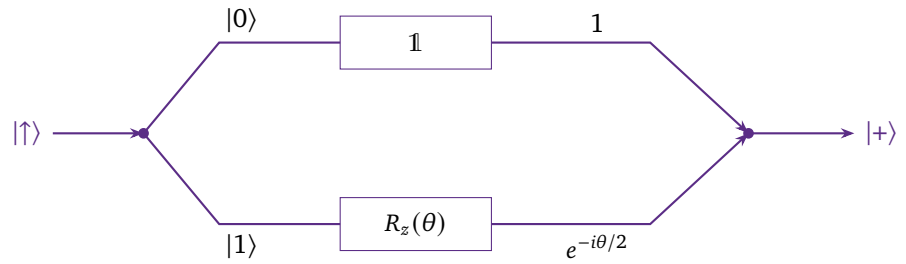


Figure 5.1: The ideal two-arm neutron experiment; only the plus output port is shown. The spin rotation acts only in path 1; the indicated factors are the relative path amplitudes before the common factor $1/\sqrt{2}$. Recombination compares them in the plus output, so a sign on one arm is observable even though a sign on both arms would be a global phase.

Rotate spin only in path 1, by the unitary $|0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes R_z(\theta)$. It gives

$$\Psi_\theta = \frac{|0\rangle + e^{-i\theta/2}\,|1\rangle}{\sqrt{2}} \otimes |\uparrow\rangle\,.$$

Recombination tests the plus path $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. The spin amplitude there is $(1 + e^{-i\theta/2})\,|\uparrow\rangle\,/2$, so

$$p_+(\theta) = \frac{1 + \cos(\theta/2)}{2}. \tag{5.4}$$

At 2π the paths cancel at this output; at 4π they add again. Rotating *both* paths by 2π instead multiplies the whole state by -1 and changes no probability. Both branches here have half-integer spin; this experiment does not superpose integer and half-integer sectors. The controlled one-arm operation is not the identity spatial rotation acting globally on the system.

The physics

Rauch et al., *Phys. Lett. A* **54**, 425–427 (1975), and Werner et al., *Phys. Rev. Lett.* **35**, 1053 (1975), observed this spinor sign using crystal neutron interferometers and magnetic precession in one beam. The ideal calculation assumes equal path amplitudes, a fixed relative phase at zero rotation, and no path-dependent loss or spin decoherence. Unpolarised spin gives the same cosine after averaging the two σ_z eigenstates. Experimental visibility and background require apparatus information beyond this model.

The physics

A conditional mass rule uses QM-II's Weyl relation. At a fixed time, take translation $U(a) = e^{-iaP/\hbar}$ and boost $V(mv) = e^{imvQ/\hbar}$. Their loop is

$$U(-a)V(-mv)U(a)V(mv) = e^{-imav/\hbar}\mathbb{1}.$$

The corresponding ordinary Galilei transformations commute in this loop. If one projective action of that ordinary group covers every admitted ray, the identity loop must be one common phase. Two different masses acquire different phases for generic av, so their coherent sum fails this premise, just as the two spin types did. This is

Bargmann's conditional mass rule, not a prohibition obtained from a mass label alone. Enlarging the group by an extra phase parameter changes the premise; see Giulini, *Ann. Phys.* **249**, 222–235 (1996), arXiv:quant-ph/9508002.

Self-check 5.5

1. Compute the plus-output probability at $\theta = \pi, 2\pi, 4\pi$.
2. Why is the measured sign consistent with the univalence restriction?

Why unitary time evolution, and which superpositions?

A continuous real group of ray symmetries has, by the cited lift, a strongly continuous unitary group. With Stone's correspondence its action is $\rho \mapsto U_t \rho U_t^\dagger$, $U_t = e^{-itH/\hbar}$, with H self-adjoint and fixed up to an energy constant. A lower bound excludes a self-adjoint time that translates energy globally. A superselection rule specifies a smaller observable algebra; its states are expectation rules on that algebra, and its pure states need not be all Hilbert-space rays.

Summary. The group premise, probability continuity and cited lift explain the structure of time evolution. Pauli's proof needs global covariance with domains, not a formal commutator. The projective rotation premise separates integer and half-integer coherent sectors. Their block algebra has unique block-diagonal density representatives, and its pure states are sector rays. A neutron's relative rotation measures a 4π path fringe without comparing the two superselected spin types.

Exercises. Try Sheet 5 in the separate *Exercise sheets* booklet (English).

Self-check answers.

5.1 1. No: the quadratic phase choice in L4 is continuous but has $c(s,t) = e^{-2ist/\tau^2} \neq 1$ until rephased. 2. $H' = H - \hbar c \mathbb{1}$, on the same domain.

5.2 1. Domain invariance makes $H\eta$ defined; the lower bound supplies $E_{\min} \leq \langle \eta, H\eta \rangle$. 2. The test-domain commutator does not imply global energy covariance. Here $D(H)$ invariance is automatic, but the energy-shift identity fails; the boundary calculation concerns $D(T)$.

5.3 The two nonzero orthogonal components would require the same scalar to equal both 1 and -1.

5.4 1. $\mathrm{diag}(1/2, 1/2)$; it is mixed on the diagonal algebra. 2. Those of E_{int} and E_{half}.

5.5 1. $1/2, 0, 1$. 2. Both arms carry half-integer spin, and only one arm is rotated.

Sources. Moretti, *Spectral Theory and Quantum Mechanics*, Theorem 12.45 pp. 714–715, Definition 12.40 p. 708; Pauli discussion pp. 827–828 (our proof is the direct lower-bound argument, with the displayed course sign convention). Superselection: §7.7 pp. 378–379, §12.3.4 p. 782; Weinberg, *The Quantum Theory of Fields I*, §2.7 pp. 81–90. Busch–Lahti–Pellonpää–Ylinen, *Quantum Measurement*, pp. 390, 395. Experimental and mass-rule citations accompany the optional applications in the full edition.